

Hormuz disruption: 40% — “Partial closure”

Bypass assumption: Conservative — EIA spare bypass. Every figure carries {source, url, retrieved, status} in the published dataset. Five pages: exposure · markets · history & structure · the 2026 ladder · method.

5.8 M b/d
STRANDED — NOWHERE TO GO

8.4 M b/d
GROSS DISRUPTED — THE HEADLINE NUMBER

2.6/2.6
BYPASS ABSORBING, M B/D

+37–69%
BRENT SHOCK BAND — ILLUSTRATIVE

208 days
STRATEGIC-RESERVE RUNWAY

+16%
SIEGE PREMIUM, 12–24 MO — PERSISTENT WHILE BARRELS STAY MISSING

4.5 M b/d
NET GAP AFTER MAX SPR RELEASE — THE TAP BINDS AT 1.3 M B/D

33 Mt/yr
LNG OFFLINE, ZERO BYPASS — GAS MULTIPLIER ×2.0–×3.1 (ILLUSTR.)

1-in-13-100
IMPLIED HULL-LOSS ODDS PER TRANSIT, FROM CURRENT QUOTES



PRODUCER	EXPORTS	STILL MOVING VS STRANDED	STRANDED	EXPOSURE
Iraq	3.2		-1.3	HIGH
Saudi Arabia	5.5		-0.7	PARTIAL
Kuwait	1.4		-0.6	TOTAL
Qatar	1.3		-0.5	TOTAL
Iran	1.5		-0.3	HIGH
Bahrain	0.2		-0.1	HIGH
United Arab Emirates	1.9		-0.0	PARTIAL

Gulf seaborne exports, M b/d, pre-crisis baselines (EIA country briefs). Per producer: stranded = exports-d – min(own pipeline spare, exports-d). Only Saudi Arabia (Petroline → Yanbu), the UAE (ADCOP → Fujairah) and Iran (Goreh–Jask) own an overland escape.

The number nobody states cleanly: Qatar’s ~77 Mt/yr of LNG — about a fifth of the world’s LNG trade — has zero overland bypass. Petroline can save Saudi barrels; nothing can save Qatari cargoes.

HOW TO READ THIS

Gross vs. net is the whole argument. Headlines quote gross throughput disrupted; the honest damage number subtracts spare overland bypass — capacity that exists and isn’t already full. EIA’s conservative spare figure is 2.6 M b/d; post-2025 Petroline-expansion commentary supports up to 5.5, so it ships as a switchable assumption, not a footnote.

The price band is illustrative by construction — real markets price expectations, OPEC spare capacity and SPR releases. History’s verdict from 45 years of Hormuz threats (p. 4): spikes mean-revert unless barrels are physically missing. This time they are.

WAR-RISK CONTEXT

Hull war-risk cover for a Gulf transit: 1–7.5% of vessel value (2026-07) vs 0.25% pre-crisis — for a \$150M VLCC, \$1.5M–\$11.3M per voyage, when cover is offered at all. Full ledger and voyage arithmetic on p. 3.

WHY HORMUZ IS DIFFERENT

The only major oil chokepoint with no sea alternative. Suez closed for eight years and the world rerouted around the Cape; here the ships have nowhere else to go (comparison, p. 4).

Importer exposure — who hurts first

An importer's loss = its Hormuz volume × d × (stranded/gross); bypass relief is pro-rata because rerouted barrels are fungible. Days of cover = strategic stocks ÷ lost supply. Sorted shortest first.

IMPORTER	VIA HORMUZ, M B/D	LOSS, M B/D	% OF ITS DEMAND	STOCKS, M BBL	DAYS OF COVER
Other Asia & Pacific ESTIMATE	6.5	-1.8	20%	90	50
India	2.1	-0.6	10%	39	67
China ESTIMATE	5.4	-1.5	9%	400	269
Rest of world ESTIMATE	0.4	-0.1	—	50	454
South Korea	1.7	-0.5	17%	240	512
Europe (OECD)	2.8	-0.8	6%	570	739
Japan	1.6	-0.4	13%	470	999+
United States	0.4	-0.1	1%	403	999+

Volumes: EIA WOTC destinations (2024) — they sum to total throughput, so importer losses sum exactly to stranded barrels (build-enforced). Stocks: IEA / ISPR / DOE; China's are state-opaque — public analyst estimates only. The asymmetry is the story: Asia holds weeks; the OECD holds a year.

RESERVES — THE TAP, NOT THE TANK REPORTED

The stock/flow runway says **208 days** — but coordinated releases max out near **1.3 M b/d** (2022 precedent: 240 M bbl over ~6 months). At today's stranded rate that covers **23%** of the gap, leaving **4.5 M b/d** uncovered. At maximum drawdown the stocks last **923 days** — the tank outlasts the crisis; the tap is the constraint.

MACRO PASS-THROUGH ILLUSTRATIVE

Public rules of thumb (+10% oil ≈ +0.2–0.4 pp CPI, -0.1–0.2 pp GDP over a year) applied to the shock band: consumer prices **+0.7–2.8 pp**; global growth **-0.4–1.4 pp**. A 2-pp global drag ≈ a typical recession's demand loss. Linear rules break down at extreme shocks — treat upper ends as directional. (IMF WEO; Fed staff ranges.)

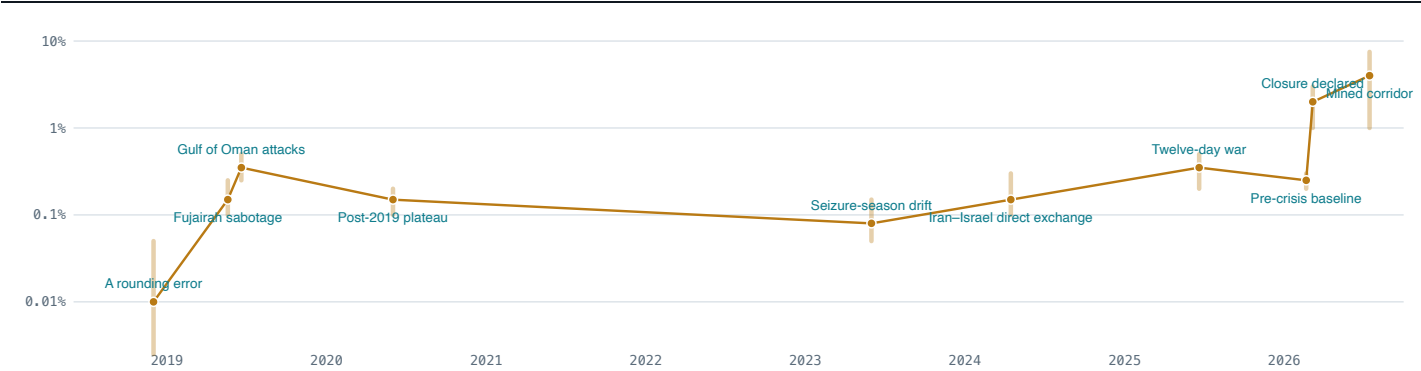
THE LNG SHOCK — NO ESCAPE EXISTS ILLUSTRATIVE

33 Mt/yr	8.3%	×2.0–×3.1
LNG OFFLINE (4.4 BCF/D)	OF GLOBAL LNG TRADE	TTF/JKM MULTIPLIER BAND
BUYER	MT/YR VIA HORMUZ	LOST AT 40%
Asia (China, India, Korea, Japan...)	60	-24
Europe	16	-6
Other	7	-3

There is no strategic reserve for gas. Europe's storage (~100 bcm, ~25% of annual demand) is a seasonal buffer measured in weeks of winter drawdown; Asian LNG buyers hold days, not months.

The empirical anchor: Russia's 2021–22 pipeline cuts removed roughly the gas-equivalent of 14% of globally traded LNG from Europe's supply — TTF peaked near 10× its pre-crisis level (≈€340/MWh in August 2022 vs ≈€35 in early 2021) before demand destruction bit. VERIFIED

War-risk insurance ledger — % of hull value per Gulf transit (log scale)



JWC listed-area circulars; Lloyd's List, S&P Global and broker commentary. Points inside quoted ranges; bars show the public low-high. From a 0.01% plumbing fee to the number that decides whether a ship sails. Note the ratchet: each crisis's "normal" settles above the last one's.

1-in-13 — 1-in-100

IMPLIED ODDS OF HULL LOSS PER TRANSIT, CURRENT QUOTES

~1-in-10,000

SAME ARITHMETIC, DEC 2018

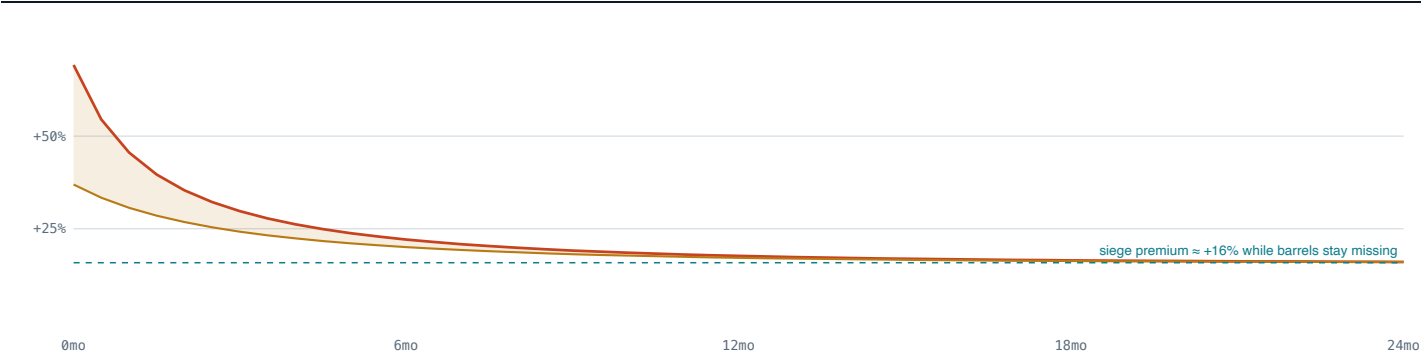
PREMIUM ≈ PROBABILITY × SEVERITY + LOADINGS → UPPER BOUNDS; AT 50% SEVERITY, DOUBLE THE 1-IN-N

Voyage economics — does the transit clear?

PREMIUM REGIME	% OF HULL	COST / TRANSIT	\$ / BBL	SHARE OF FREIGHT	VERDICT
Feb 2026 “pre-crisis”	0.25%	\$0.3M	\$0.15	4%	CLEARs
Current — safer vessels	1%	\$1.2M	\$0.60	15%	CLEARs
Current — mid	4%	\$4.8M	\$2.40	60%	MARGINAL
Current — high-risk	7.5%	\$9.0M	\$4.50	113%	UNECONOMIC

\$120M VLCC, 2 M bbl cargo, \$4.0/bbl freight (public broker commentary; crisis chartering has spiked rates 3x+). One-way transit; round trips can double the premium. Above ~100%, the underwriter — not the charterer — decides whether the ship sails.

Dynamic price path — the spike decays; the siege premium stays



$\epsilon(t) = \epsilon_0 + (\epsilon_\infty - \epsilon_0)(1 - e^{-t/\tau})$, $\epsilon_\infty = 0.35$, $\tau = 6$ months **ILLUSTRATIVE** — assumes the disruption persists; expectations, SPR releases and GDP feedback excluded. Substitution and demand destruction tame the spike, but converge to a persistent premium — not to zero — while barrels stay missing.

Event study — Brent after Hormuz threats (1/5/30 trading days)

EVENT	DATE	+1D	+5D	+30D
Tanker War escalates off Kharg ESTIMATE	1984-05-24	+1.0%	+2.0%	-4.0%
Bridgeton mine strike VERIFIED	1987-07-24	-2.8%	+0.2%	-10.2%
Operation Praying Mantis VERIFIED	1988-04-18	+3.6%	+1.8%	-1.5%
Iran closure threat (sanctions) VERIFIED	2011-12-27	-1.6%	+3.7%	+1.1%
Fujairah sabotage VERIFIED	2019-05-12	+1.0%	+3.2%	-11.3%
Gulf of Oman attacks VERIFIED	2019-06-13	+2.6%	+1.9%	+9.7%
Abqaiq-Khurais strike VERIFIED	2019-09-14	+11.7%	+6.5%	-4.0%
Soleimani strike VERIFIED	2020-01-03	+3.0%	-0.7%	-13.8%
Iran strikes Israel directly VERIFIED	2024-04-13	-2.4%	-5.5%	-13.0%
Israel-Iran twelve-day war VERIFIED	2025-06-13	+7.3%	+13.5%	+1.7%
2026 Hormuz crisis begins VERIFIED	2026-02-28	+8.3%	+34.2%	+70.9%

EIA Europe Brent spot (RBRTE) via FRED public mirror DCOILBRETEU, daily; windows computed by the fetch pipeline (pre-1987 rows predate the daily series and remain estimates). The pattern: threat spikes mean-revert within weeks — even Abqaiq, the largest single supply disruption in oil-market history, round-tripped in a month — **unless barrels stay missing**. The 2026 row is the counterexample that proves the mechanism: +71% at 30 days and climbing.

Chokepoint anatomy — why Hormuz is different

CHOKEPOINT	OIL, M B/D	SEA ALTERNATIVE	BYPASS, M B/D	CLOSURE RECORD
Straits of Hormuz	20.9	NONE — NO WAY OUT BY SEA	2.6	Never fully closed through 45 years of threats — until 2026. The first actual closure is the current crisis.
Straits of Malacca	23.7	Sunda / Lombok straits +1-3d	0.4	Never closed; piracy-era harassment at worst. Rerouting costs days, not barrels.
Suez Canal + SUMED	9.2	Cape of Good Hope +10-14d	2.5	Closed 1956-57 and 1967-1975 (eight years); Ever Given 2021 (six days). The world rerouted and lived.
Bab el-Mandeb	8.6	Cape of Good Hope +10-14d	—	2023-25 Houthi campaign roughly halved transits — traffic diverted, cargoes still arrived.
Panama Canal	0.9	Cape Horn / Suez routings +5-13d	—	2023-24 drought cut transits ~30% — products rerouted at a freight cost.

EIA WOTC reference volumes. Every other chokepoint has an escape — a longer sea lane, a parallel pipeline, or both. Suez closed for eight years and the world rerouted. Hormuz's first real closure is a different kind of event.

Reopening runway — why d→0 is not day zero

FRICTION	PERSISTENCE	ANCHOR
Price premium	2-6 weeks	Threat premiums mean-revert within weeks once barrels flow — unless barrels stay missing. Abqaiq 2019: +14.6% one-day spike fully round-tripped in under a month despite 5.7 M b/d offline. VERIFIED
War-risk insurance	4-18 months	Underwriters normalize on renewal cycles and JWC reviews, not headlines — premiums stay multiples of the old baseline for months to years. After the 2019 spiral, Gulf rates held ~10x their April-2019 level well into 2021. ESTIMATE
Mined corridor	3-9 months	A swept convoy corridor takes months; certified full clearance takes years. Mines outlast every ceasefire that laid them. 1991 Gulf War: 1,200+ mines; coalition clearance operations ran for over two years. VERIFIED

The road to closure — 2025 → 2026

2025-06-13	Israel strikes Iran — the twelve-day war VERIFIED Strikes on nuclear and military targets; Iran answers with missile barrages. Tankers begin GPS-jammed, zig-zag transits. Flows: Transits continue; loadings dip · Price: Brent +~7% day one
2025-06-22	US strikes Fordow; Majlis votes to close the strait VERIFIED B-2s hit Fordow, Natanz, Isfahan. Iran's parliament votes (non-binding) to close Hormuz — the decision rests with the SNSC, which doesn't act. The strait stays open. Flows: Self-deterrence: Iran's own exports transit the same water · Price: Brent touches ~\$81 then collapses -7% on ceasefire news (24 Jun)
2026-02-28	The 2026 crisis begins REPORTED US and Israeli strikes on Iranian leadership and military targets. Iran's response makes 45 years of closure threats operational for the first time. Flows: Immediate transit hesitancy; insurers pause quotes · Price: +8.3% day one, +34% in a week, +71% in a month (daily spot series) — the first Hormuz spike that did not mean-revert
2026-03-02	IRGC declares the strait closed REPORTED VHF broadcasts warn 'no ship is allowed to pass'; transit permitted only for Iran-approved vessels. Mining of the central corridor and attacks on transiting ships follow. Flows: Tanker transits collapse toward -70% · Price: War-risk quotes move from 0.25% to 1-3% of hull within a week
2026-04-11	US mine-clearing operations begin REPORTED Two destroyers transit to 'set conditions' for de-mining; blockade operations against Iranian naval logistics run April-May. Flows: Escorted convoys only; ~150+ vessels damaged or on fire to date (reported) · Price: High-risk hull quotes reach 7.5-10%
2026-06-15	Persian Gulf Strait Authority stands up ESTIMATE Multinational escort-and-transit authority begins organizing convoys through swept corridors. ~80 mines still reported in the central corridor as of mid-July. Flows: Partial, escorted resumption; ~70% of normal transits still missing · Price: Premiums stay 4x-30x pre-crisis

THE MODEL

stranded(d)	= T·d - min(spare_bypass, T·d)
ΔP/P	≈ share / ε, ε ∈ [0.08, 0.15], capped
price path	: ε(t) = ε₀ + (ε∞-ε₀)(1-e ^{-(t/τ)})
SPR offset	= min(max_release_rate, stranded)
producer_i	: exports_i·d - min(pipeline spare_i, ·)
importer_i	: imports_i·d · stranded/gross
LNG mult	: 1 + (lng_lost/global_trade) / ε_gas
implied odds	: P(loss) ≤ premium / severity
voyage	: share = hull·p / (cargo·freight)
macro	: (ΔP%/10)·[coeffs], one-year

PROVENANCE LEGEND

verified — checked against the named public source at build time · **reported** — live-crisis public reporting · **estimate** — derived/approximate · **illustrative** — model assumption · **brief-baseline** — pending a fresh primary-source check. If a figure is an estimate, it says so.

SOURCES

EIA World Oil Transit Chokepoints (retrieved 2026-07-22); EIA country briefs, STEO and daily Brent · IEA oil security & gas market reports; GIIGNL · ISPR, DOE, KNOC/METI stock disclosures · JWC listed-area circulars; Lloyd's List, S&P Global and broker commentary · CRS R45281; US Navy histories · 2026-crisis public reporting · IMF WEO / Fed staff rules of thumb · Natural Earth. Per-figure citations with URLs and retrieval dates ship in the dataset and render in the site's Methodology panel.

INTENDED USE

Decision-support for shipping, energy and policy analysis from 100% public data. No real-time vessel positions, no navigational data, nothing that could serve as a targeting aid. Incident positions are approximate; vessel-behavior content is pattern-level.

Interactive version, all scenarios: **chokepoint.analyticadss.com**
Scenario deep link: <http://localhost:5199/?intro=skip&brief=1#v=map&d=40&b=eia&y=2026&e=11>
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